

Radiative Geometry: Matter, the First Law, and the Failure of Statics on Substrates That Pass the Area Gate

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The companion paper of this campaign measured the area gate, the requirement that the states counted for a region scale with the region's boundary, and found one mechanism class that passes it, deterministic local dynamics read over spacetime regions. This paper asks what geometry such a substrate can carry, and answers with exact measurements. Matter on the maximal-entropy linear substrate is exactly invisible to every local window, and the one source class that produces a finite distinguishability field destroys exactly one bit per time step, with a field placed by the algebra of the dynamics rather than by distance. A local first law holds exactly on held-out source patterns, with flux, temperature, and entropy defined independently, once the flux counts the events crossing into the spacetime region. The Newtonian limit is rejected. The discrete analogue of the Poisson equation fails, no monotone flank supports a flux reading, and the information deficits of separated sources cancel instead of adding, so the black-hole benchmark that would follow is forbidden by its own design bar. Two discovery campaigns then attacked the failed bar and closed with it intact. A structured vacuum removes the invisibility and yields a genuinely graded field, but the field's curvature lives on the causal front, and escape to infinity yields an exact static limit that is flat. Moving to two dimensions leaves finite-field substrates flat at every distance, while a real-amplitude wave substrate makes superposition deviations decay cleanly with source separation and then sends its own static field toward zero as the transient radiates away. On both substrate classes that pass the area gate, every spatial gradient of distinguishability is radiative. The area law requires zero entropy production, zero entropy production makes relaxation purely radiative, and purely radiative relaxation leaves no static field for a Poisson structure to live in. A constraint experiment then measures the one escape the chain permits. A declared local constraint of Gauss type, realized as lattice electromagnetism with charges entering only through the constraint, carries an exactly static, graded, and exactly superposing distinguishability field whose potential solves the lattice Poisson equation. Under a demotion ruling recorded before the run, that result does not support entropy as the source of geometry. The constraint sources the geometry and distinguishability registers it exactly, locating the missing ingredient of the failed bar in a local conservation law rather than in entropy. A last gate asks whether such a constraint can emerge, and finds that under generic local deformation the Gauss structure collapses rather than deforms, so constraints are declared or absent, never emergent in this class. Nothing in this paper bears on physical gravity.

I. INTRODUCTION

Entropic accounts of gravity assume that the entropy associated with a region of space is proportional to the area of the region's boundary, and derive gravitational dynamics from that input together with a local thermodynamic relation [1–3]. The companion paper of this campaign treated the assumed proportionality as a gate and measured it in models where every bit is countable [11]. The verdict of that paper was a classification. Every declared mechanism with a positive bulk density of entropy fails the gate before it is built, and the one mechanism class that provably passes is deterministic local dynamics, with the counted region read as a patch of spacetime history rather than a snapshot of space. Determinism concentrates all randomness on the initial data, so the states of a spacetime patch are bounded by causal-cone data whose size scales with the patch perimeter, for every rule in the class and with nothing tuned.

That result opened a question rather than closing one. A mechanism that passes the counting requirement is only the first half of an entropic account of geometry. The second half needs matter to deform the substrate, needs a local thermodynamic law to hold with its quantities independently defined, and needs the resulting field to obey the structural consequences of the Newtonian limit, none of them inserted. This paper reports the campaign's measurements of that second half on the substrate class the gate selected, and the outcome is exact on both sides.

The positive side is a lawful thermodynamics. A local first law holds exactly on every held-out source pattern, with the flux, the temperature, and the entropy change measured by three constructions that make no reference to one another. The relation holds precisely when the flux counts the events crossing into the spacetime region, which is the ordering the thermodynamic-gravity literature assumes rather than derives.

The negative side is the geometry. Matter on the maximal-entropy substrate has no local field at all, and the one source class that leaves any trace produces a field arranged by the algebra of the dynamics rather than by distance from the source. The deficits of separated

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sources cancel instead of adding. Every structural test of the Newtonian limit fails, and the failure survives two discovery campaigns that were mounted specifically to defeat it. The campaigns removed two obstructions, an ensemble artifact and a topology artifact, and then located the remaining obstruction precisely. On every substrate measured, the gradients of distinguishability travel. They live on causal fronts and in transients, and whatever static limit exists is either flat or zero. The organizing claim of this paper is a three-step chain stated and measured exactly in the sections below. The area law requires zero entropy production, zero entropy production makes relaxation purely radiative, and purely radiative relaxation leaves no static field for a Poisson structure to live in.

The chain is the result for substrates without constraints, and a constraint experiment measured the one escape it permits. A local conservation law of Gauss type, declared as a constraint rather than discovered, holds an exactly static and graded field through the same kind of reversible dynamics. A ruling recorded before that run demotes the claim such a result can earn. The constraint sources the geometry and the distinguishability field registers it, so the missing ingredient of the geometric bar lies in conservation structure, which entropy alone never supplied. A last gate then asks whether such a constraint can emerge from generic local dynamics, and the measured answer is that it cannot, so the track's closing question is why nature has constraints at all.

The scope of every claim is the declared model class. The measurements are exact computations in toy constructions whose every ensemble, source, and instrument was declared before the runs. No claim is made about physical spacetime, about real gravitation, or about the correctness of any entropic-gravity proposal.

II. THE SUBSTRATE AND THE INSTRUMENTS

The companion paper's results are used here in summary form and are not re-reported [11]. Its instrument layer validated conditional-entropy and relative-entropy measurements against exact closed forms. Its gate run classified constraint, gauge, and access-restriction mechanisms as volume-law failures in the primary count, measured an exact split between the independence count and the correlation count of one Gauss-law ensemble, and proved two structural statements, an entropy-density lemma forcing any passing ensemble to zero bulk entropy density and a causal-cone theorem giving every deterministic nearest-neighbor rule perimeter-scaling patch counts. The mechanism run bound those theorems to measured rule-90 patch counts with a bulk-noise audit. Everything below builds on that substrate class.

The vacuum of the first three experiments is the maximal linear substrate. A ring of 257 binary cells evolves for 60 steps under the elementary rule 90, in which each cell updates to the sum modulo two of its two neighbors, from independent uniform initial data. Because the rule

is linear over the two-element field, the distribution of any window of the spacetime history is uniform on the image of a linear map, so every comparison between a matter ensemble and the vacuum ensemble is exact subspace arithmetic. A window comparison lands in one of three classes. The two supports may be equal, so the relative entropy is zero. The matter support may be nested strictly inside the vacuum support, so the relative entropy is a finite whole number of bits, the dimension deficit. The matter support may escape the vacuum support, so the relative entropy is infinite. All three classes are decided by rank computations with no estimator and no sampling anywhere.

Matter is a rule substitution. Cells of a declared source region update by rule 150, which adds the cell's own value to the sum of its neighbors, inside the rule-90 vacuum. Sources of widths 1, 3, 5, and 7 cells were declared. A rule substitution carries no potential, no distance function, and no coupling constant, and the anti-circularity contract of the track design binds every run.

III. EG-2: MATTER IS EXACTLY INVISIBLE LOCALLY

The first question is whether matter deforms the fiber measure at all, and the measured answer on this substrate is that locally it does not. All numbers in this section are from the committed evidence record of the matter run.

Every window outside the defect's causal cone carries the identical subspace, so the causality null holds exactly for every source. Inside the cone the invisibility is total. Every single cell, every two-by-two patch, and every row segment of lengths 16, 32, and 64 at every one of 25 scanned positions classifies equal to vacuum for every source width, 75 window comparisons per source with not one exception. Matter has no local field on this substrate. The reason is the ensemble. Uniform initial data makes every local marginal already maximal, so there is nothing local left to distinguish.

Visibility arrives only globally, and how it arrives depends on the source width in a way the algebra dictates. Sources of widths 1, 5, and 7 remain exactly equal to vacuum on every proper segment up to length 256 and escape support only at the closed ring of all 257 cells. The vacuum ring carries 256 bits because rule 90 forces one global parity constraint on each row, and the defect frees that constraint, giving the matter ring 257 bits. The entire effect of these sources is one global all-or-nothing bit.

The width-3 source is the unique exception, and it is a sink. It destroys rank instead of freeing it, at the exact rate of one bit per time step, so the global deficit after 60 steps is exactly 60 bits, the duration itself. This is the only source class with a finite distinguishability field anywhere. The field appears at window length 160 and grows with window length, with deficits of 11, 27, 43, 51, 56, 59, and 60 bits at lengths 160 through 256, about half

TABLE I. The held-out patterns of the first-law run. Events inside count injections within the spacetime patch, naive events count all injections including those in the patch’s causal past, and the entropy change is an exact rank difference in bits. The relation $\Delta S = T \Delta Q$ with T frozen at one bit per event holds with zero residual on every row under inside accounting, and the two rows where the accountings differ are the ones that identify which flux the law selects.

Pattern	ΔQ inside	ΔQ naive	ΔS (bits)
dilute, 2 events	2	2	2
dilute, 4 events	4	4	4
moderate, 8 events	8	8	8
all in causal past, 4	0	4	0
mixed, 2 inside 2 past	2	4	2
stacked column, 12 events	12	12	12

a bit of deficit per added cell on the growing windows. Its spatial arrangement is a period-64 self-similar pattern. In the fixed-length position scan the finite values repeat every 64 cells with the profile 3, 7, 11, 9, 5, 1 in the first period and 2, 6, 10, 9, 5, 1 in the later ones. The pattern is set by the automaton’s algebraic self-similarity and not by distance from the source, which is the first sign of the geometric failure measured in Sec. V.

The reverse-engineering audit for this experiment is structural rather than numerical. For any potential one can define an entropy whose gradient reproduces the corresponding force, so a source class must be certified not to be a potential in disguise. A positive reweighting of a measure cannot move its support, while a rule substitution provably does move the support of a uniform-on-subspace measure, as the escapes above witness. The declared source class is therefore not potential-equivalent by construction, and the audit required by the track design passes without a fit.

IV. EG-3: THE LOCAL FIRST LAW HOLDS EXACTLY

The first law experiment asks whether $\Delta S = T \Delta Q$ holds with the three quantities defined independently, which is the relation the entropic-gravity lineage consumes as an input [1, 3]. On this substrate it is an output. All numbers in this section are from the committed evidence record of the first-law run.

The three definitions never reference one another. The flux ΔQ counts declared noise injections inside the spacetime patch, pure event bookkeeping with no entropy anywhere in its definition. The temperature T is the patch-entropy response to a single training injection, measured once before the held-out sweep and frozen at exactly one bit per event. The entropy change ΔS is an exact rank difference between the injected and vacuum patch ensembles. The vacuum patch itself carries 120 bits, exactly the perimeter from the causal-cone bound prescribes for its size.

Table I shows the six held-out patterns. Dilute, moderate, and stacked interior patterns satisfy the relation exactly, with the residual zero to rounding. The structural finding is in the two rows where the accountings disagree. An injection before the patch’s time span is absorbed exactly, because the maximal ensemble is already uniform on every reachable set, so a detector reading a later time slice is provably blind to it. The all-past pattern accordingly gives zero entropy change against a positive naive flux, and the mixed pattern holds under inside-only accounting while the all-events accounting overpredicts. The Clausius relation is an output on this substrate precisely when the flux is the one crossing into the spacetime region. That ordering, a first law referred to the flux through a causal boundary, is what Jacobson’s construction assumes when it places the relation on local horizons [1], and here it emerges from the accounting rather than being imposed. The caveat is adjacent and structural. The absorption argument leans on the maximality of the vacuum ensemble, so this exactness is a property of the measured substrate, not a theorem about all ensembles.

V. EG-4: THE NEWTONIAN LIMIT IS REJECTED

The geometric experiment takes the only field-bearing source class, the width-3 sink, and tests the structural consequences of the Newtonian limit on held-out configurations. All numbers in this section are from the committed evidence record of the geometry run. An honest process note belongs with them. A draft verdict written before the run anticipated a different failure pattern, contradicted the measured items, and was discarded in favor of verdicts computed from the numbers. The evidence record carries only the computed verdicts.

The Poisson item fails. The discrete Laplacian of the measured field is as large far from the source as at it, reaching magnitude 6 at positions far outside the source against magnitude 3 at the source cells, so nothing resembling a source-localized second difference exists. The Gauss item fails. The field profile is the period-64 algebraic pattern of Sec. III, and no monotone flank exists anywhere on it to carry a flux reading.

The superposition item fails, and the manner of failure is the section’s sharpest measurement. Two separated width-3 sinks produce a global deficit of exactly zero, at both tested separations, against the additive expectation of 120 bits. Three sinks produce a deficit of exactly one bit. The deficits are dimensions of subspace intersections, and they compose by the algebra of the two-element field, not by addition. This is not weak nonlinearity to be tamed in a weak-field limit. It is destructive interference of information loss, and the linearity item fails with it. The interior item is unevaluable, because windows between the sources and windows far outside both classify equal, so the field has no interior-versus-exterior structure at the tested scale.

One law does survive, and its exactness matters for what follows. Each isolated sink loses exactly one bit per step at either tested position, 60 bits in 60 steps. The substrate supports an exact local law of information loss. What it refuses to support is any spatial organization of that loss into a potential.

VI. EG-5: THE GATE CLOSES BY ITS OWN BAR

The black-hole benchmark of the track design may not run unless the matter, first-law, and geometry experiments all pass, because a horizon analogy computed on top of a failed geometry would be theater. The geometry experiment rejected the Newtonian limit, so the benchmark is forbidden by construction and the planned arc closes. Its closing shape is the tension this paper is about. The counting half of the entropic program wants zero entropy production, which is what buys the area law and the exact first law. The geometry half wants a graded local field, and the measurements above show the same determinism that delivers the first half destroying the second. The two discovery campaigns reported next were declared, run, and closed in the attempt to break that tension without insertion.

VII. THE FIRST DISCOVERY CAMPAIGN: A STRUCTURED VACUUM

The campaign declaration, recorded in the track document before any run, diagnosed the invisibility of Sec. III as an artifact of the maximal vacuum rather than of determinism. Uniform initial data leaves every local marginal full, so nothing local can be distinguished, while the causal-cone bound and the first law are ensemble-independent. The declared move keeps the deterministic dynamics and gives the vacuum structure, product Bernoulli initial data with bias 0.2. Window distributions are then exactly computable by Walsh analysis, in which the expectation of each parity functional of the window is determined by the size of its support, so every field value is an exact Kullback-Leibler divergence with no sampling. The instrument was validated against brute-force enumeration to 8.3×10^{-17} before any field was read, and the causality null is exact with the ring wrap accounted for. All numbers in this section are from the committed evidence record of the discovery campaign.

Round one removed the invisibility obstruction. A single rule-substitution source in the structured vacuum produces a genuinely graded real-valued field, shown in Fig. 1, a plateau of 0.812 bits inside the cone with smooth symmetric monotone flanks rising over about eight cells, where the maximal vacuum gave exactly zero everywhere. Two declared items pass immediately, gradedness and monotone flanks for a Gauss reading. Two fail informatively. The discrete Laplacian does not concentrate at

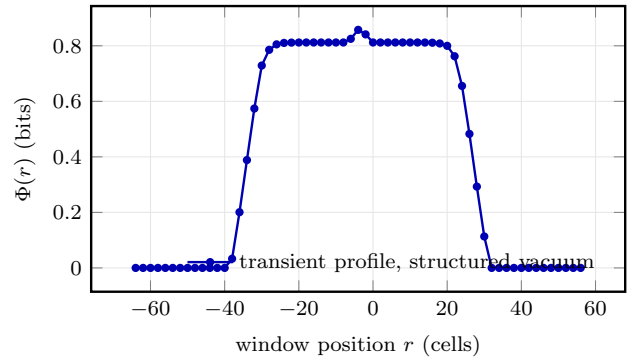


FIG. 1. The round-one field of the first discovery campaign, the relative entropy of the eight-cell window at position r against the structured vacuum, read from the committed discovery record. The plateau sits at 0.812 bits inside the causal cone with smooth symmetric monotone flanks, against exactly zero everywhere on the maximal vacuum of the matter run. The small excess near the source is the single point-like feature the escaped static limit later retains. Every coordinate is read from the committed evidence record by the committed figure script.

the source, reaching 0.098 on the causal front against 0.049 at the source, so the field's curvature lives on the front. And the superposition deviation, between 3.6 and 6.3 percent across the declared source strengths, does not shrink as the source weakens, because a rule substitution is a quantized order-one perturbation. The two-element field offers no continuous weak-field knob.

Round two showed the ring has no static limit. With the source left on until the causal front wraps the ring and self-collides, the profile flattens to a near-uniform 0.226 bits with spread 3.6 percent and keeps drifting between late times by 23 percent. The round-one gradient was entirely front-borne, and once the front has nowhere to go the gradient dies with it.

Round three escaped to infinity and found exact statics, but flat. On a lattice too large for any wrap at the tested times, the near-source profile converges exactly under time doubling, with successive relative changes of 1.0×10^{-3} and then 0.0. This is a true static limit, reached by radiating the switch-on transient away to infinity, which is how a reversible system relaxes without producing entropy and therefore without harming the area law. The converged field, however, is flat. It sits at 0.812 bits at every scanned offset with a single point-like bump at the source and spread 5.3 percent, carrying no spatial gradient at all.

The campaign verdict, computed from these measurements, is that the geometric bar is not passed and the obstruction chain is fully mapped. Local invisibility was an ensemble artifact, removed by vacuum structure. The missing static limit was a topology artifact, removed by escape to infinity. What remains is structural and twofold. The static distinguishability field of this substrate class carries no spatial gradient, because all gradient is radia-

tive and lives on causal fronts, and dynamics over the two-element field admits no continuous weak-field parameter.

VIII. THE SECOND CAMPAIGN: DIMENSION, THEN AMPLITUDE

The second campaign attacked the two remaining obstructions with a declared falsifiable hypothesis per round, each recorded in the track document before its run.

Round one tested dimensional dilution. The hypothesis held that the flat statics of the first campaign are a one-dimensional artifact, since one-dimensional radiation cannot dilute, while two-dimensional radiation thins geometrically with distance and might leave a graded static response. The substrate is the two-dimensional rule in which each cell becomes the sum modulo two of its four nearest neighbors, with the defect rule adding the cell's own value at source sites, structured Bernoulli initial data, and the same exact Walsh instrument with functionals computed by backward evolution. The hypothesis is refuted. The two-dimensional axis profile is flat again, 0.0611 bits at every distance with spread 7×10^{-6} , the diagonal shows only parity oscillation, and superposition deviations sit at one half of scale at every separation. The refutation sharpens the diagnosis. The obstruction is not dimension but the finite field itself. Influence under linear rules over the two-element field is parity path-counting, which is zero or one and never small, so no finite-field linear substrate can dilute with distance in any dimension. Geometric dilution requires amplitudes that decay.

Round two moved to real amplitudes, with the physics motivation declared openly in advance. The substrate is the two-dimensional discrete wave equation in leapfrog form, reversible, deterministic, local, and linear over the reals, with a declared site stiffness of 0.2. The vacuum is independent standard Gaussian initial data, and matter is an impedance defect, a site of different stiffness, still a rule substitution with no potential anywhere. Since the static limit of the wave equation is the Poisson equation, this is the substrate where the Poisson item has its best chance, and saying so before the run is what keeps the test honest. Window marginals are exact Gaussians whose covariances are computed by adjoint evolution, and the covariance instrument was validated against dense covariance evolution to 1.1×10^{-15} , with causality exact at unit wave speed. The first two instrument iterations found the impedance signal at or below the float noise floor of the covariance calculus, fields near 10^{-5} bits with hundredfold isotropy violations diagnosed as ill-conditioning of highly correlated neighboring cells, an instrument lesson recorded in the evidence record. The third iteration reads a strong-contrast scatterer, stiffness 0.05 in the 0.2 vacuum, through decorrelated stride-two windows, and the field becomes well-measured, up to 5×10^{-3} bits with isotropy within the profile spread. The campaign's second breakthrough is in Fig. 2(a). Superposition deviations de-

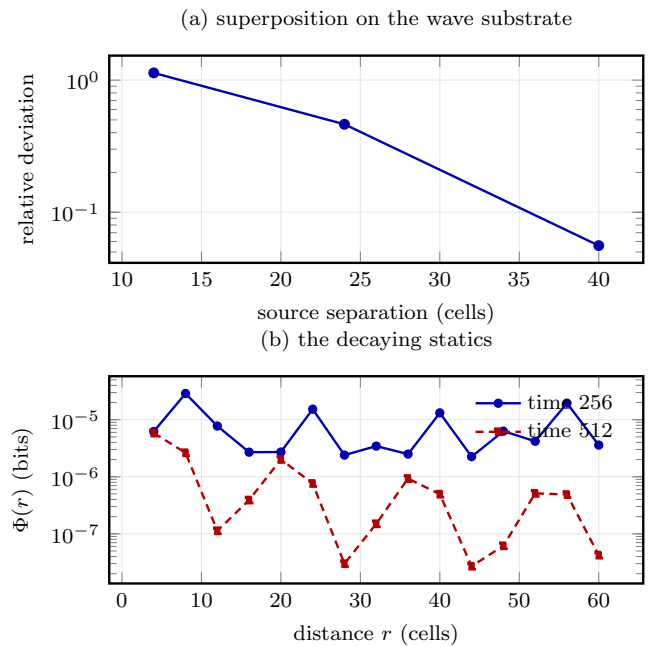


FIG. 2. The second discovery campaign on the real-amplitude wave substrate. Panel (a) shows the superposition deviation of two impedance scatterers falling cleanly with their separation, the continuous weak-field behavior no finite-field substrate could provide. Panel (b) shows the near-source field at doubling times on wrap-free lattices. The field decays with time at every distance, so the static limit is consistent with zero. Every coordinate is read from the committed evidence records by the committed figure script.

cay cleanly with source separation, 1.13 to 0.46 to 0.056 at separations 12, 24, and 40. Distance works as the continuous weak-field knob, which no finite-field substrate could provide. What round two could not reach at times 48 and 96 is the static regime itself, since two-dimensional waves relax with slow algebraic tails and the profiles remained transient-dominated.

Round three ran the long statics batch, reading the axis profile at doubling times 256 and 512 on lattices too large for any wrap, shown in Fig. 2(b). The static regime is not reached, and the direction of failure is itself the answer. The near-source field decays with time, from about 3×10^{-5} bits at time 256 down to between 10^{-7} and 6×10^{-6} at time 512. As the switch-on transient escapes to infinity, nothing persistent remains near the source. The static limit of the wave substrate's distinguishability field is consistent with zero.

The campaign closing verdict, computed from the three rounds, is that the geometric bar stands and the obstruction is now measured on both substrate classes that pass the area gate. On the finite-field substrate the static field converges exactly but is flat. On the real-amplitude wave substrate the static field decays toward zero as the transient clears. The campaign's positive yields also stand, the structured-vacuum mechanism for local visibility, the escape-to-infinity route to reversible relaxation, and dis-

tance as a continuous weak-field knob on real-amplitude substrates.

IX. CONSTRAINT-LOCKED STRUCTURE: THE ESCAPE ROUTE MEASURED

The closing verdict of the second campaign left one escape route, a persistent local structure that survives its own relaxation. Physics keeps exactly one standard mechanism of that kind, a local constraint of Gauss type tying field flux to conserved charge on every time slice. A constraint-anchored field cannot relax away, because every configuration the dynamics can reach satisfies the same constraint. The campaign had already met this mechanism’s information signature once, since the Gauss-law ensemble of the companion paper carried an exact area-law mutual information across every cut while failing the gate’s primary count [11], and nothing in this section claims that failure repaired.

A claim demotion was ruled in the track document before the run. If a Gauss constraint is declared at the base level, the static Poisson structure of the mean field follows from the constraint mathematics and not from entropy, so no positive result here can support the reading that entropy sources geometry. The honest claim the experiment can earn is that constraints source geometry and distinguishability registers them exactly. What remains nontrivial and measurable is whether the relative-entropy field is static under the declared dynamics, graded with a falloff measured rather than assumed, exactly superposing with the correct charge scaling, and Gauss and Poisson consistent on the measured field.

The declarations are these. The substrate is two-dimensional lattice electromagnetism on a torus of side 128, with the electric field on edges, the magnetic field on plaquettes, and the standard local curl-pair dynamics, which are linear, reversible, local, and preserve the per-vertex divergence of the electric field exactly. The vacuum is the identity-covariance Gaussian on the fields conditioned on zero divergence at every vertex, and it is exactly stationary because the transverse sector rotates isotropically while the longitudinal sector is frozen by the dynamics. Matter is a conserved charge assignment, a plus and a minus unit charge separated by 64 sites, entering only through the constraint value. Every window marginal is an exact conditional Gaussian, with means from one Fourier solve and covariances from Green-function lookups, and the instrument was validated against dense conditioning on a small lattice to 5.4×10^{-16} in the mean and 3.1×10^{-16} in the covariance. The audit records that the constraint is a local per-vertex statement about incident edges and that no radial function, no potential, and no Poisson equation is inserted anywhere in the declarations. All numbers in this section are from the committed evidence record of the constraint run.

All eight declared items pass. The distinguishability field is exactly static under the deterministic dynamics,

with a drift of 2.0×10^{-18} over 400 steps, the anti-radiative witness every previous substrate failed, because the conditional mean is an exact fixed point of the curl pair and the conditioned isotropic ensemble is stationary. The field is graded and monotone with a measured log-log slope of -2.275 against the two-dimensional Coulomb expectation of -2 . A first run on a 64 lattice with separation 32 measured -2.715 , the steepening of a dipole read at ranges comparable to its own separation, and the recorded remedy was the larger lattice, not a refit. The field-level object superposes exactly, with deviation 4.2×10^{-16} , and the relative entropy scales exactly quadratically in the charge, with the doubled-charge ratio at 4.0 against the predicted 4.0. On the measured mean field the Gauss law holds to 8.9×10^{-16} and the curl vanishes to 1.5×10^{-18} , so the field’s potential solves the lattice Poisson equation.

The demotion ruling governs the reading of every number above and belongs beside them. This result does not support entropy sourcing geometry. The Poisson structure is a derived consequence of the declared local constraint, so the constraint sources the geometry and the relative-entropy field registers it exactly. What the measurement establishes, in one construction and with the campaign’s exactness, is the resolution of the track’s three-way tension. A local conservation law tying matter to boundary flux is the one measured mechanism that carries persistent graded structure through reversible dynamics, and it is the missing ingredient of the geometric bar, located rather than circumvented.

X. THE EMERGENT-CONSTRAINT GATE (EG-6)

After the constraint experiment the question of reopening the black-hole benchmark was ruled on, and it stays closed. Constraint-inherited passes with no horizon, no surface-gravity analogue, and no acceleration-dependent temperature would produce exactly the ambiguous scorecard the design philosophy exists to prevent, so the track’s successor gate asks a sharper question instead, whether a constraint of Gauss type can emerge rather than be declared.

The question has an exact operational form. For linear local dynamics $\dot{x} = Ax$ a frozen functional is a left null vector of A , and Gauss structure is a translation-invariant family of local left null vectors, which is exactly what Maxwell’s per-vertex divergences are. The gate measures the dimension of the frozen-functional space and the locality of its members over a declared family of local translation-invariant deformations of the curl pair, perturbing both blocks of the dynamics. A first one-block deformation falsely showed the family surviving, because the divergence-curl identity of the untouched electric-field update carried it unaided, and that instrument lesson is recorded. All numbers below are from the committed evidence record of the emergence run.

Measured on a torus of side 8, the frozen space collapses

from 66 to 14 dimensions at deformation 10^{-3} and stays at 14 identically through deformation 0.3. The residual of the Gauss family against the frozen space jumps from 2.2×10^{-15} to 0.884 at the same step. The fourteen survivors are non-local by an exact criterion. A local frozen functional supported in a window exists precisely when the frozen projector's principal submatrix on that window has an eigenvalue equal to one, and the maximum radius-one window eigenvalue is 0.2188 at every nonzero deformation against the Maxwell control's exact 1.0000.

The verdict, computed from the numbers, is the designed negative. Gauss structure within this class is an isolated algebraic point, declared or absent, never approached by local deformation, so the emergence of constraints requires a mechanism outside generic local linear dynamics.

XI. EVERY GRADIENT IS RADIATIVE

The measurements of the two campaigns assemble into one statement. On both constraint-free substrate classes that pass the area gate, every spatial gradient of distinguishability is radiative, carried by transients and causal fronts, and nothing graded survives the static limit. The finite-field substrate converges to a flat static field. The wave substrate's static field vanishes. The chain behind this is three exact steps. The area law requires zero entropy production, which is the entropy-density lemma of the companion paper read as a constraint on dynamics. Zero entropy production makes relaxation purely radiative, because a reversible system can shed a transient only by transporting it away, never by absorbing it. And purely radiative relaxation leaves no static field for the Poisson structure to live in, as both campaigns measured. The chain is the campaign's result for constraint-free substrates.

The contrast with ordinary statics locates the tension in familiar territory. A dissipative system relaxes to a graded static profile by pushing entropy into a bath, in the manner the quantum Brownian motion literature makes precise [5], and it is exactly that entropy production which the area law forbids. The transients that carry the gradients here play the role the decoherence literature assigns to the environment as a record carrier [4]. The information about the source is not destroyed but exported to infinity, and once exported it grades nothing locally. Within this model class the two halves of the entropic-gravity program pull in opposite directions. The counting half wants zero entropy production, and the geometry half wants a graded static field that zero entropy production forbids.

The open problem this chain appeared to leave behind, a persistent local structure that survives its own relaxation without the answer being inserted, is answered by the constraint experiment of Sec. IX under its demotion ruling. The missing ingredient of the geometric bar is located, a local conservation law tying matter to boundary

flux, and entropy alone never supplied it. On constraint-free substrates every gradient radiates away, while the one measured mechanism that holds a gradient still is a constraint that sources the geometry itself, with the relative-entropy field registering it exactly. Any future claim that entropy alone produces Poisson structure on an area-law substrate now has a measured counterexample program to answer. The emergence gate then closes the remaining loophole from the other side, since within generic local linear dynamics a Gauss structure cannot arise by deformation and is declared or absent. The track's final question is therefore no longer whether entropy sources geometry but why nature has constraints at all, and that question leaves this campaign's methods for other ones.

XII. RELATION TO PRIOR WORK

Four literatures border the results, and each border is stated with what a specialist there would already know.

The rigidity measured at the emergence gate is a small finite-dimensional shadow of the consistent-deformation program for gauge theories, which classifies the possible deformations of a gauge structure cohomologically and finds gauge symmetries rigid under broad conditions [6]. A specialist in that program would consider the non-genericity of local constraint structure known in spirit and proved in far greater generality. What is the paper's own is the falsification-gate treatment, an exact measured collapse of the frozen space with the window criterion supplying a numerical locality witness against a Maxwell control.

The radiative-statics obstruction is a measured cousin of the critique that entropic forces struggle to reproduce conservative static potentials. Visser showed that demanding a conservative static force field of an entropic account places severe constraints on its entropy and temperature functions [7], and Carroll and Remmen dissect which entropy an entropic derivation actually needs and where its assumptions enter [8]. The difficulty itself is therefore known. What is the paper's own is the three-step chain from area law to absent statics measured exactly on declared substrates, with the static limit computed rather than argued.

The constraint-registration reading connects to entanglement entropy in gauge theories, where the entropy assigned to a region depends on how the Gauss constraint is handled at the region's boundary [9]. That constraints shape region entropies is known there in the quantum setting. What is the paper's own is the inversion thesis assembled with exact numbers, a constraint sourcing an exactly static geometry while the relative-entropy field registers it, on a substrate where every quantity is computable in closed form.

The emergence question has an affirmative answer in condensed matter, where frustrated systems realize emergent gauge fields as low-energy effective constraints [10]. That answer does not contradict the emergence gate, be-

cause its mechanism lies outside the gate’s class. Energetic projection into a constrained subspace at low temperature is not kinematic freezing in generic local linear dynamics, so the gate’s negative delimits where emergence can come from rather than denying that it occurs.

XIII. WHAT THIS PAPER ESTABLISHES

Table II registers the claims with their labels. Read together they complete the arc the companion paper opened. The substrate class that passes the counting gate carries a lawful and exactly measured thermodynamics, refuses every structural test of Newtonian geometry, and does so for a reason the discovery campaigns located precisely. The refusal is not an artifact of ensemble, topology, dimension, or alphabet, each of which was removed or refuted in turn. It is the radiative character of relaxation without entropy production. The one measured escape is a declared local conservation law, and under the recorded demotion that escape sources the geometry rather than earning it from entropy.

XIV. METHODS AND REPRODUCIBILITY

All constructions extend the campaign’s published instrument stack [11, 12] under the track’s anti-circularity contract. The forbidden-input list excludes any inserted potential, any distance function, any entropy-area proportionality, and any temperature defined through the law it verifies, and every ensemble, source class, probe, and bar was declared in the track document before the corresponding run. Both discovery campaigns and each round of the second campaign carry dated declarations

recorded before any run, including the declared falsification conditions of their hypotheses, and the constraint experiment’s claim demotion was likewise recorded before its run. Verdicts are computed from the measurements after the runs. The geometry run enforces this discipline’s value in one recorded lesson, a draft verdict written before the data contradicted the measured items and was replaced by the computed verdicts the evidence record now carries.

Every result-bearing run writes an append-only JSON evidence record containing the complete declared parameter set, outcome arrays, and a canonical hash of the record body. The records for this paper are `eg2-matter.json`, `eg3-first-law.json`, `eg4b-discovery.json`, `eg4c2-wave.json`, `eg4d-gauge.json`, `eg6-emergent.json`, `eg4-geometry.json`, `eg4c-mini.json`, and `eg4c3-statics.json` in the campaign repository, nine records all carrying the exploratory label of the campaign’s evidence-grading scheme, and the highest label any claim here earns is demonstrated in model. The finite-field instruments are exact rank and Walsh computations with no sampling, validated against brute-force enumeration, the wave-substrate instrument is exact Gaussian covariance evolution, and the constraint-substrate instrument is exact conditional-Gaussian calculus validated against dense conditioning, with the entropy background of the comparison classes as in the standard treatment [13]. There are no free parameters anywhere in the analysis chain. Every number and every figure coordinate in this paper is read from those committed records or is an exact closed form derived in the text, with the figure tables generated deterministically by a committed script. A reducer may summarize records but may not rewrite them.

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- [1] T. Jacobson, Thermodynamics of spacetime: the Einstein equation of state, *Phys. Rev. Lett.* **75**, 1260 (1995), arXiv:gr-qc/9504004.
 - [2] G. Bianconi, Gravity from entropy, *Phys. Rev. D* **111**, 066001 (2025), arXiv:2408.14391.
 - [3] P. Dorau and A. Much, From quantum relative entropy to the semiclassical Einstein equations, *Phys. Rev. Lett.* **136**, 091602 (2026), arXiv:2510.24491.
 - [4] W. H. Zurek, Decoherence, einselection, and the quantum origins of the classical, *Rev. Mod. Phys.* **75**, 715 (2003).
 - [5] A. O. Caldeira and A. J. Leggett, Path integral approach to quantum Brownian motion, *Physica A* **121**, 587 (1983).
 - [6] G. Barnich and M. Henneaux, Consistent couplings between fields with a gauge freedom and deformations of the master equation, *Phys. Lett. B* **311**, 123 (1993), arXiv:hep-th/9304057.
 - [7] M. Visser, Conservative entropic forces, *J. High Energy Phys.* **10** (2011) 140, arXiv:1108.5240.
 - [8] S. M. Carroll and G. N. Remmen, What is the entropy in entropic gravity?, *Phys. Rev. D* **93**, 124052 (2016), arXiv:1601.07558.
 - [9] H. Casini, M. Huerta, and J. A. Rosabal, Remarks on entanglement entropy for gauge fields, *Phys. Rev. D* **89**, 085012 (2014), arXiv:1312.1183.
 - [10] S. Sachdev, Topological order, emergent gauge fields, and Fermi surface reconstruction, *Rep. Prog. Phys.* **82**, 014001 (2019), arXiv:1801.01125.
 - [11] A. H. Bond, An exact area gate for entropic accounts of geometry: volume laws, count splitting, and the causal-determinism mechanism, companion paper of this campaign (2026).
 - [12] A. H. Bond, Three sources of observed entropy in reversible hidden dynamics: mixing, projection multiplicity, and inaccessible degrees of freedom, companion paper of this campaign (2026).
 - [13] T. M. Cover and J. A. Thomas, *Elements of Information Theory*, 2nd ed. (Wiley-Interscience, Hoboken, 2006).

TABLE II. Claim register for the post-gate stage of the entropic-geometry track. Demonstrated in model means an exact measurement in a declared construction with append-only evidence records and verdicts computed from the measurements. Closed by design means a consequence of a preregistered bar, not a measurement. The EG-4d rows are read under the claim demotion recorded in the track document before that run. Every number traces to a committed evidence record named in Sec. XIV.

Result	Claim	Label
EG-2, invisibility	matter on the maximal vacuum is exactly equal to vacuum in every local window for every source width, and widths one, five, and seven alter only one global parity bit at the closed ring	demonstrated in model
EG-2, sink law	the width-three source is the unique rank-losing sink, exactly one bit per step, and its finite field is a period-64 algebraic pattern set by the dynamics, not by distance	demonstrated in model
EG-3, first law	$\Delta S = T \Delta Q$ holds exactly on all held-out patterns with flux, temperature, and entropy independently defined, and holds precisely under inside-the-region flux accounting	demonstrated in model
EG-4, Newtonian limit	Poisson, Gauss, superposition, and linearity all fail, with deficits of separated sinks composing by two-element field algebra, zero for two sinks against an additive expectation of 120	demonstrated in model
EG-5, benchmark	the black-hole benchmark may not run on a failed geometry, so the planned arc closes with the area mechanism and the first law as positives and the geometry as the negative	closed by design
EG-4b, visibility	a structured Bernoulli vacuum restores a graded field with a 0.812-bit plateau and monotone flanks, so the invisibility was an ensemble artifact	demonstrated in model
EG-4b, no statics	the field's Laplacian lives on the causal front, the ring has no static limit, escape to infinity converges exactly but flat, and the two-element field has no continuous weak-field knob	demonstrated in model
EG-4c, round one	dimensional dilution is refuted, the two-dimensional finite-field profile is flat at every distance, so the obstruction is the finite field, not dimension	demonstrated in model
EG-4c, round two	on the real-amplitude wave substrate, superposition deviations decay cleanly with source separation, so distance is a continuous weak-field knob	demonstrated in model
EG-4c, round three	the wave substrate's near-source field decays with time on wrap-free lattices, so its static limit is consistent with zero	demonstrated in model
EG-4d, statics	the constraint-conditioned field of lattice electromagnetism is exactly static under the curl dynamics, drift 2.0×10^{-18} over 400 steps, graded and monotone with measured falloff slope -2.275 against the Coulomb expectation of -2	demonstrated in model
EG-4d, derived Poisson	the field superposes exactly and scales quadratically in the charge, and its potential solves the lattice Poisson equation as a derived consequence of the declared constraint, which sources the geometry that distinguishability registers	demonstrated in model
EG-6, emergence	under local translation-invariant deformation of the curl pair the frozen-functional space collapses from 66 to 14 dimensions at deformation 10^{-3} , the Gauss family's residual jumps from 2.2×10^{-15} to 0.884, and every survivor is non-local by the exact window criterion, so Gauss structure is declared or absent, never emergent in this class	demonstrated in model
Synthesis	on both measured constraint-free substrate classes passing the area gate, every spatial distinguishability gradient is radiative, the three-step chain from area law to absent statics holds exactly, and the one measured escape is a declared local conservation law	demonstrated in model
Scope	nothing in this paper bears on physical gravity, and no physics claim is made	non-claim